The Global Metric Proxy Fallacy

An Empirical Audit of APY Failure Modes in the G2F Maize Panel

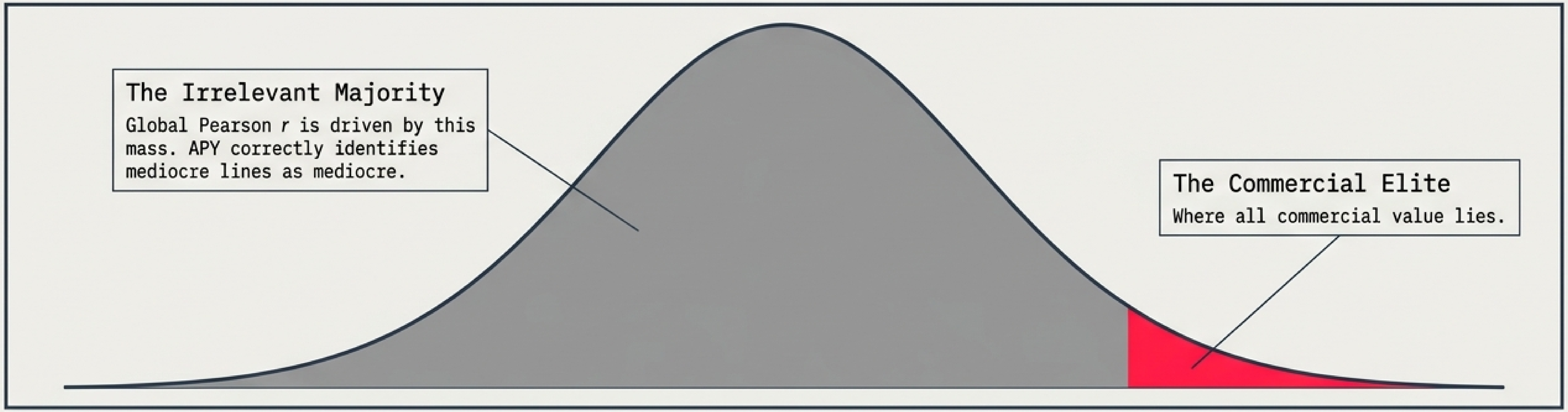
TARGET_DATASET:
Genomes To Fields (G2F)
Maize 2014–2023

SOLVER_COMPARISON:
APY Approximation vs.
MPDOK Exact G-BLUP

EVALUATION_METRIC:
Top-1% Elite
Precision @ k=21

Key Takeaway: A conclusive demonstration of how a widely accepted global correlation metric obscures catastrophic selection failures in structured crop populations.

The Vanity Metric: Why Global ρ ≈ 0.98 is a Distraction



The Literature's Claim

Validation protocols measure Pearson r across the ENTIRE population.

In homogeneous populations (e.g., Holstein cattle), this yields $r > 0.99$.

The Breeder's Reality

Commercial programs discard the middle 90%. Success is dictated entirely by the right tail.

A metric dominated by discarded lines is mathematically blind to the misranking of the top 1%.

DATA PROOF: In the real-world G2F Maize dataset, global Pearson r drops to +0.21 on $\hat{u}$, and precision at the Top-1% is a catastrophic 4.8%.

The Test Subject: 10 Years of Real Hybrid Yields

Data sourced directly from the public Genomes to Fields (G2F) Genotype × Environment Prediction Competition.

```
Hybrid observations:          161,534
Environments (Loc × Year):    272 (2014–2023)
Named Genotyped Inbreds:      2,191
SNPs (MAF > 1%):              48,580
```

Phenotype Modeled: General Combining Ability (GCA) for grain yield via sparse least squares.

GRM Effective Rank: ≈ 126

```python
# Build sparse design matrix and solve for GCA
obs = np.arange(n_obs)
env_col = df['Env'].map(env_idx).values
```

```python
y = df['Yield Mg ha'].values
print(f'Design matrix: {X.shape}, {X.nnz:,} nonzeros (sparse
      fraction: {X.nnz / (n_obs * (n_env+n_gca))})')
```

```
Design matrix: (161534, 2463), 484,602 nonzeros (sparse fraction:
1.22e-03)
lsqr converged in 1.2s | residual norm: 744.9523 | stop reason: 2
```

Population Structure:

The panel spans six distinct heterotic groups (Iowa Stiff Stalk, Lancaster Sure Crop, BSSS, CIMMYT). The limited inter-group recombination is the mathematical tripwire that will break APY.

Elite Selection Failure: Missing 20 of 21 True Elites

At n_core = 200, APY misses 95% of the true top-1% inbred lines identified by exact G-BLUP.

True Elites Buried by APY			APY False Positives		
TX736	**True Rank #1** (Highest GCA at +2.98) →	APY Rank #1,086	F42	**APY Rank #1** →	True Rank #1,683
PHR03	**True Rank #2** →	APY Rank #1,112	MBNIL_B040	**APY Rank #3** →	True Rank #1,944 (Bottom 11%)
B73	**True Rank #3** (Universal Reference Genome) →	APY Rank #1,089			
OH43	**True Rank #13** →	APY Rank #195			

TAKEAWAY: APY's top selection is literally the 1,683rd-best inbred in reality. The failure is absolute.

The Mechanism: 91× Inflation and Sign Reversal

Step 1: Inflation | Raw Solution Vector $\alpha = (G+\lambda I)^{-1}y$

The Trap: APY uses numerical inversion of a near-singular core submatrix.	**The Result:** Extreme inflation of non-elite α values.

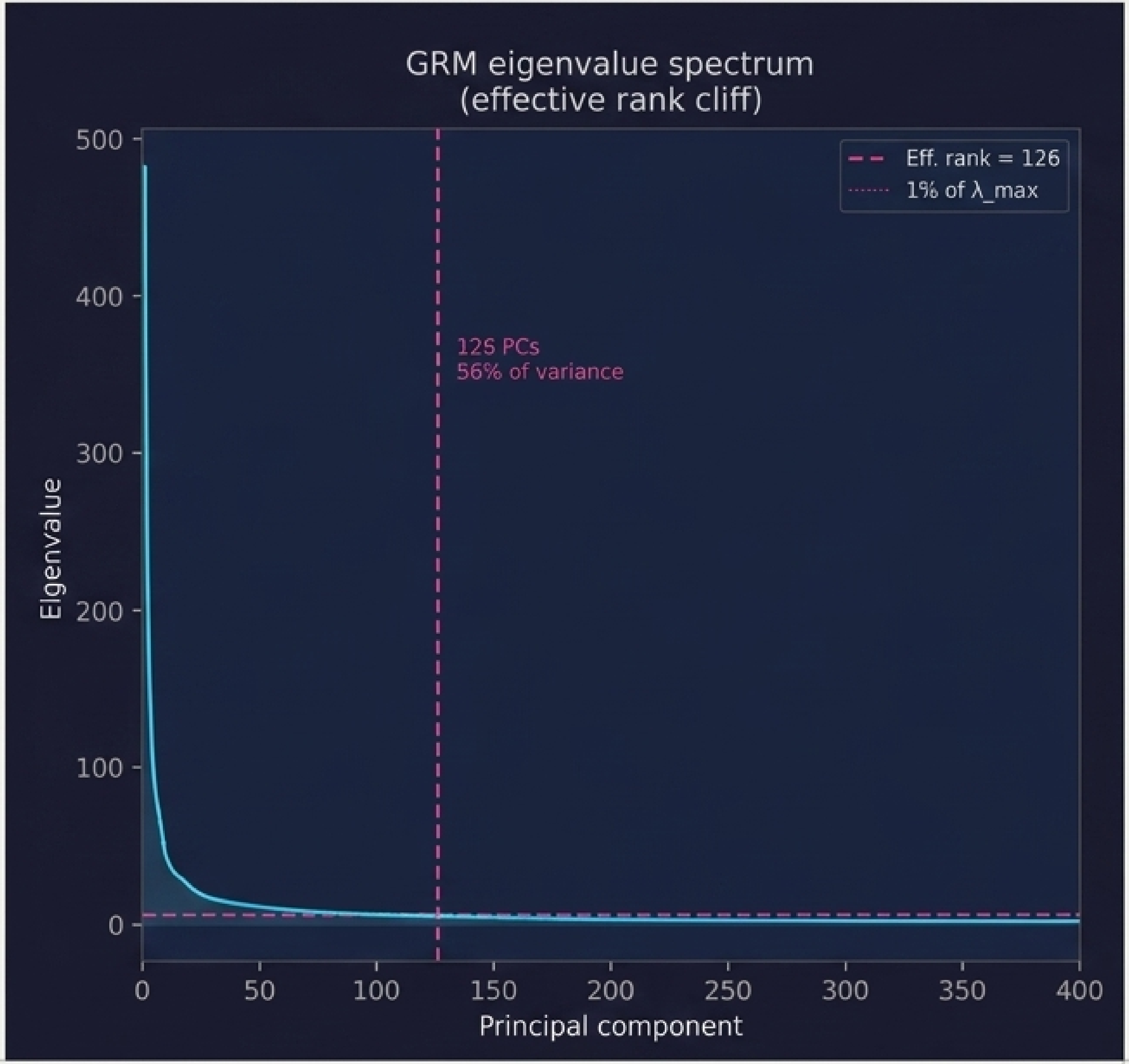
```
Data Proof: F42 (APY #1) α inflated by 91× (+35.6 exact vs +3,263 APY)
Data Proof: MBNIL_B040 inflated 33×
```

Step 2: Sign Reversal | Estimated Breeding Values $\hat{u} = G \cdot \alpha$

The Trap: G has large negative inter-group off-diagonals (reflecting divergent heterotic ancestry).	**The Result:** Inflated false-positive α values map into massive, sign-revt! penalties for true ISS-group elites.

```
Data Proof: TX736 (True #1, Exact û = +1.48) receives an APY û of −305.3 Mg/ha.
A complete sign reversal and 205× magnitude error.
```

The Structural Tripwire: Effective Rank vs. Core Size



The Cliff

The **first 126 eigenvalues** account for 56% of panel variance. The remaining 2,065 are effectively zero.

The mathematical reality: the **effective rank** of this 2,191-line panel is **≈ 126**.

The APY Pathology Box

CONDITION: n_core = 200 > effective rank = 126.

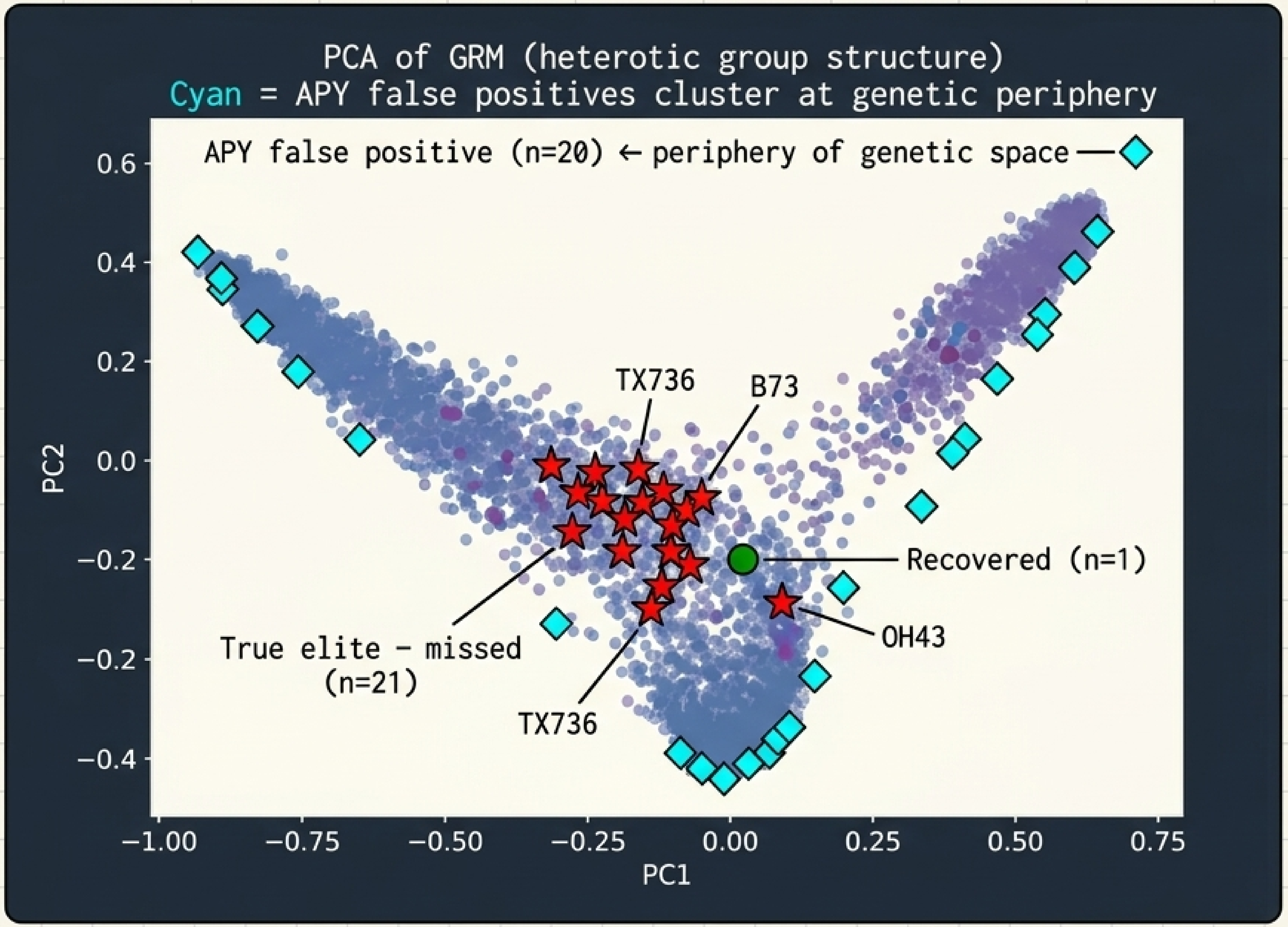
→

CONSEQUENCE: Any core submatrix (G_pp) drawn from this panel will contain near-zero eigenvalues.

→

CONCLUSION: APY's Sherman-Morrison-Woodbury inversion becomes numerically unstable **by construction on multi-group panels.**

The Peripheral Outlier Map: The Boundary-Candidate Trap



Insight Callout

Observation

APY false positives (cyan diamonds) perfectly trace the extreme edges of the genetic landscape, far from the dominant germplasm.

Mechanism

The SMW inversion amplifies α most severely for lines whose genetic neighborhoods are poorly represented in the core.

The mathematical instability literally pulls its false-positive selections from the isolated edges, burying the true central elites.

The Anatomy of a Total Rank Inversion

Deconstructing the Scatter

The Red Vertical Scatter

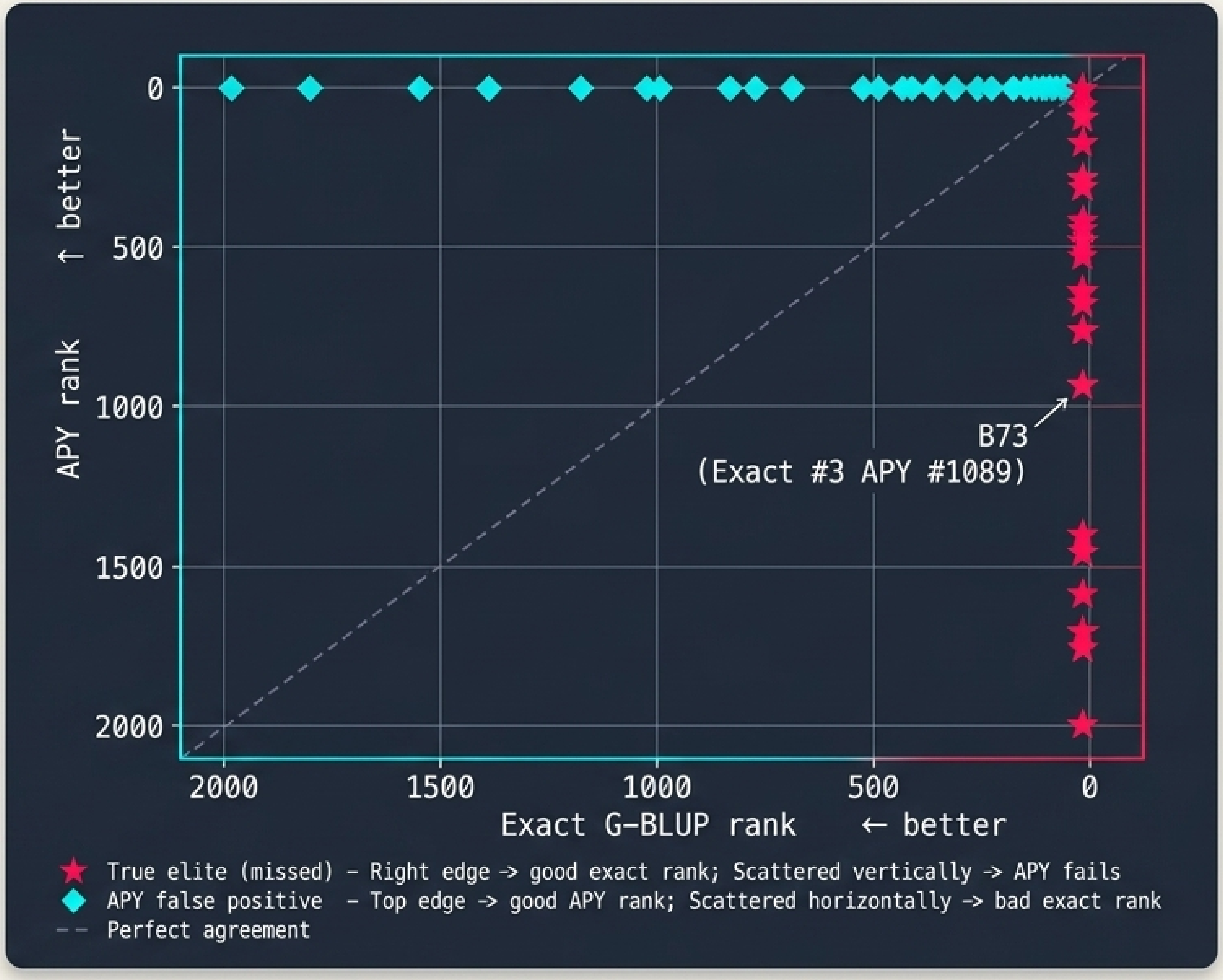
True elites (Red Stars) sit on the far right edge (Top 1% Exact) but are scattered vertically across the entire Y-axis (APY ranks from 1 to 2000).

The Cyan Horizontal Scatter

APY Top Picks (Cyan Diamonds) sit on the top edge (Top 1% APY) but span the entire X-axis horizontally, pulling deep from the bottom 7% of the real distribution.

The Ultimate Failure

Note the B73 data point. Exact rank #3, but pushed down to APY rank #1,089.



NotebookLM

Core-Set Instability: Selection by Russian Roulette

Testing 100 random core draws at n_core = 200 reveals structural instability.

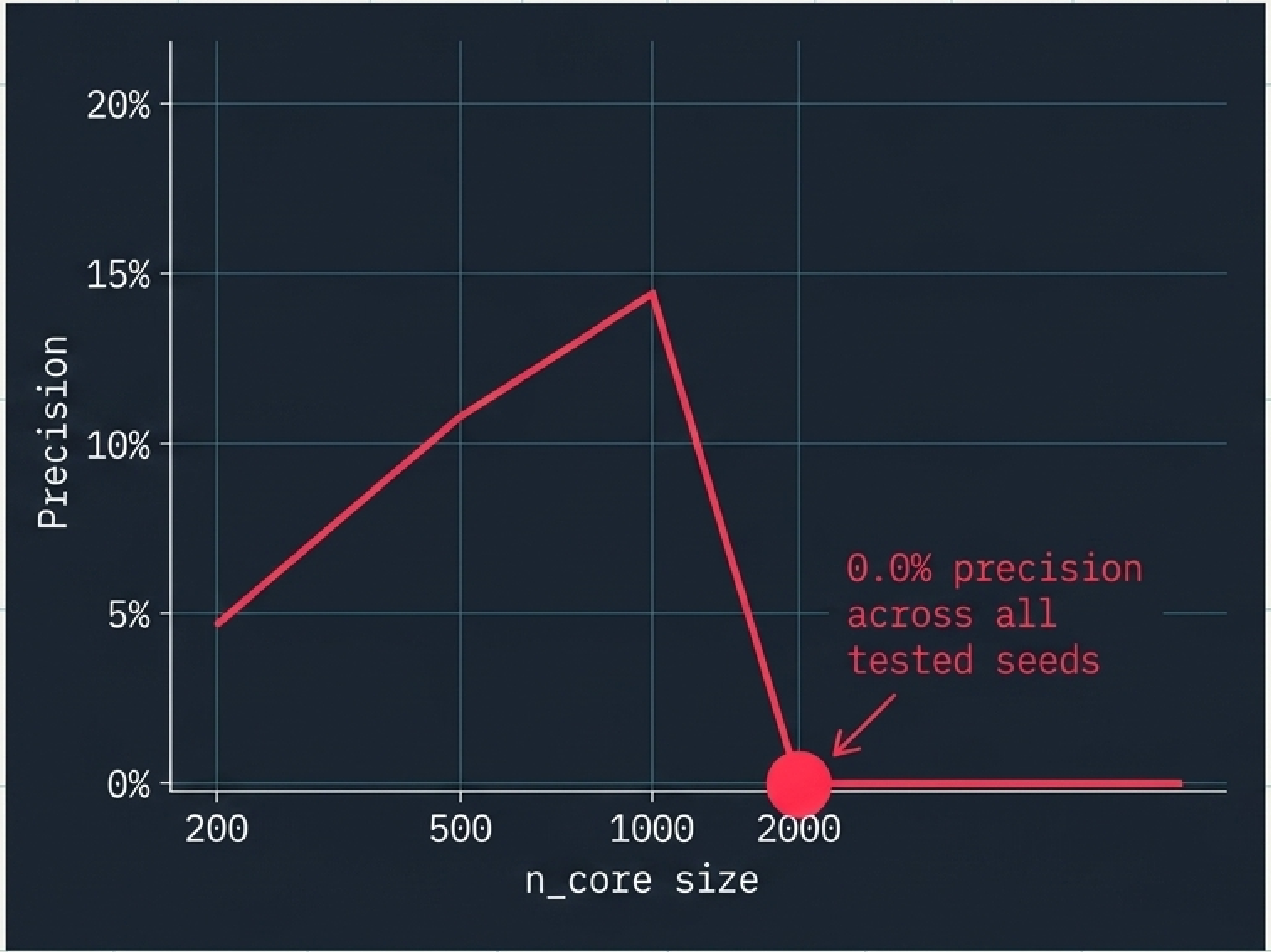
Mean Top-1% Precision	Standard Deviation	Best-Draw Precision	Worst-Draw Precision
4.8%	4.3% (CV = 89%)	19.0%	0.0% (zero true elites recovered)

The Structural Explanation

The 0% worst-case is not a fluke or a bug. It happens naturally when the random draw happens to concentrate core lines in a subset of heterotic groups, leaving ISS-group elites entirely in the non-core block.

APY performance is entirely at the mercy of the random seed.

The Paradox of Scale: A 0% Precision Collapse at n_core = 2,000



Counter-Intuition

The Assumption:

If APY covers 91% of the panel (2,000 of 2,191 lines), shouldn't it converge to the exact answer?

The Reality:

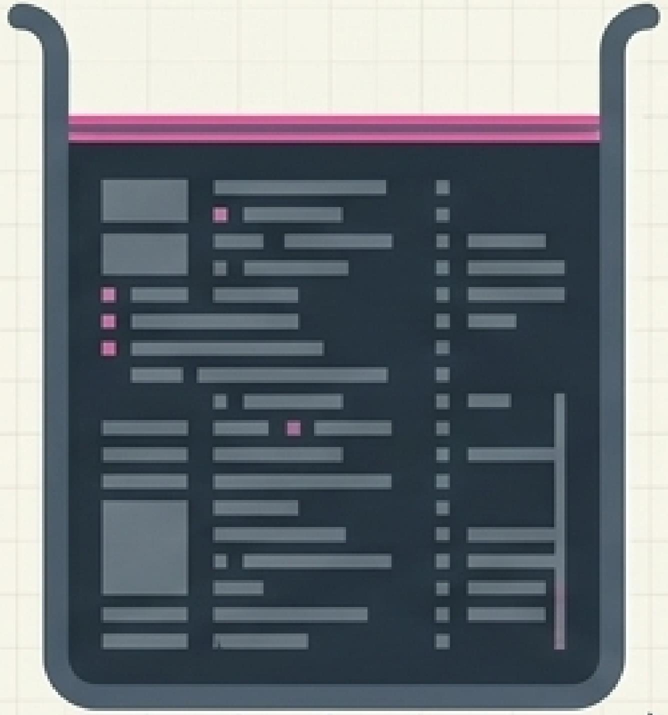
In a homogeneous population, yes.

In a structured crop panel, scaling up the core randomly triggers an inescapable algebraic trap.

This is a structural breakdown, not low-rank approximation error.

The Mechanism of Collapse: The Ψ Vanishing Act

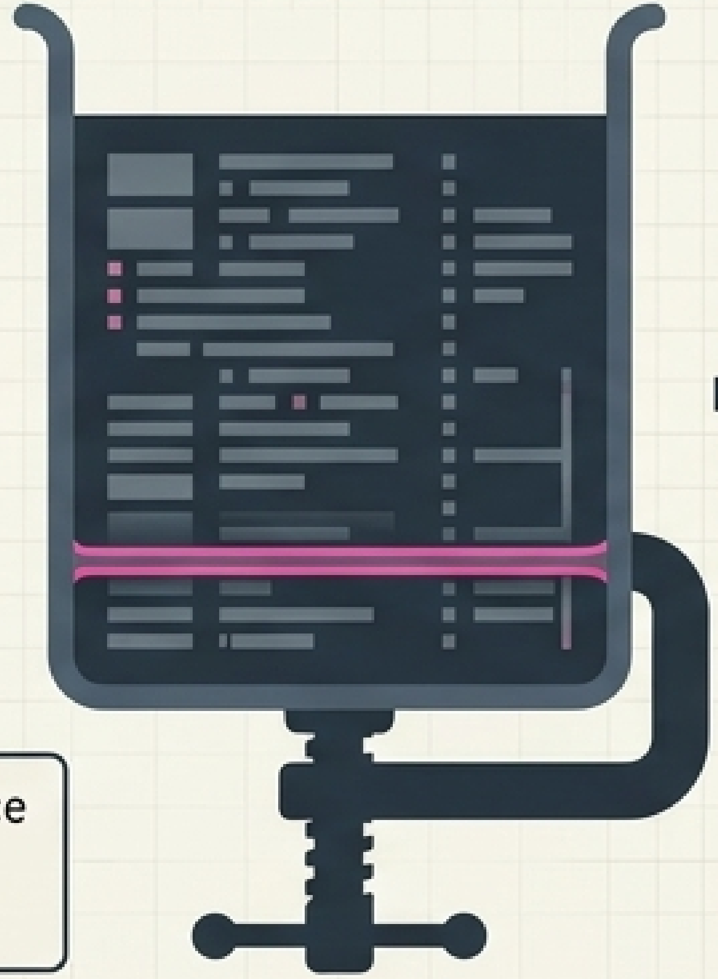
Step 1: Extreme Over-conditioning



Residual variance (Ψ_{ii}) drives to zero or below.

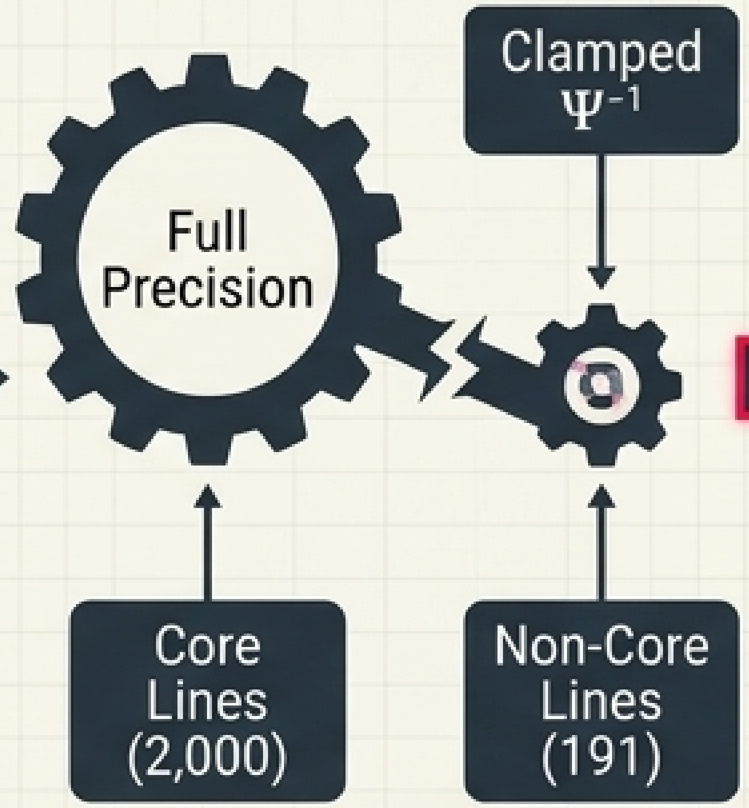
With 2,000 lines in the core, the genealogical overlap computationally fully explains the remaining 191 non-core lines.

Step 2: Artificial Damping (Clamping)



To stop Ψ^{-1} from exploding, APY clamps it to an artificial minimum. This uniform clamp ignores actual genomic variance structure.

Step 3: Scale Discontinuity



Core lines are **inverted** at **full precision**.
The 191 non-core lines are inverted through an arbitrarily clamped Ψ^{-1}, subjecting them to severe shrinkage toward the mean.

Step 4: The Elite Lockout

If true elites (like ISS-group TX736) land in the 191 non-core block, their α values are crushed toward zero by clamping.

They are mathematically locked out of top rankings.

The Diagnostic Audit: Defending the Results (Part 1)

These results are extreme. **We executed 5 independent double checks to definitively rule out coding bugs.**

Audit Matrix

✓	**Check 1: Sort Direction** **Audit:** Is argsort(-alpha) really descending?	**Result:** TX736 is rank 1 globally, N is global minimum. Sorting is flawless.
✓	**Check 2: Identity Map** **Audit:** Does y_{gca} perfectly match the named dictionary map?	**Result:** All spotlights align exactly. GCA-to-GRM alignment is mathematically intact.
✓	**Check 3: Solve Residual** **Audit:** Does $(G+\lambda I)\cdot\alpha_{exact} \approx y_{GCA}$?	**Result:** Relative RMS is 1.99×10^{-7}. Exact solve is numerically precise.

The Diagnostic Audit: Defending the Results (Part 2)

Audit Matrix

✓	**Check 4: Non-Monotonic n_core Scaling** **Audit:** Does APY recover linearly toward exact solve as n_core increases?	**Result:** No. Across 3 seeds, n_core=1000 produces worse max error than n_core=500. This confirms numerical instability of near-singular G_pp, refuting smooth rank-truncation. Convergence only happens identically at n_core = N.
✓	**Check 5: Subpopulation Fingerprint** **Audit:** Where are the false positives coming from?	**Result:** WGS_2018-2019 dataset is enriched 3.0× in APY false positives relative to its panel frequency. **Conclusion:** Seed 42 randomly over-sampled PHN11/PHW65 lines, causing APY to structurally inflate their α values via singular inversion. This is a predictable mathematical signature.

The Real-World Cost: 97.5% Loss in Genetic Gain

Evaluated under the Breeder's Equation: $\Delta G = i \cdot r \cdot \sigma_A$.

	APY (n_core=200)	MPDOK Exact
Top-1% Precision	4.8%	100%
Gain Efficiency S_{APY} / S_{exact}	2.5%	100%
Effective Selection Intensity i_{eff}	0.103	2.680
True Elites Missed	20 of 21	0 of 21

THE BOTTOM LINE: APY delivers just **2.5%** of the theoretically achievable genetic gain per selection cycle on this panel. **97.5%** of selection intensity is vaporized by ranking disruption.

The Necessary Transition: MPDOK Exact Solve

The Structural Pathology

APY is structurally pathological for admixed diversity panels, multi-group crops, or any collection with distinct ancestral pools where n_core > effective rank. Continuing to use APY means misranking commercial elite lines every breeding cycle.

The Exact Performance Matrix

Algorithm: LU Factorisation with FP64 iterative refinement.

Hardware: Single RTX GPU.

GRM Build Time: 1.6 seconds.

Exact G-BLUP Solve Time: 0.95 seconds.

Elite Recovery: 21 out of 21.

FINAL TAKEAWAY: There is no longer a computational excuse for relying on unstable approximations. The exact solution computes in **under one second.**